x = g(x) Iteration

Exam code: 9MA0

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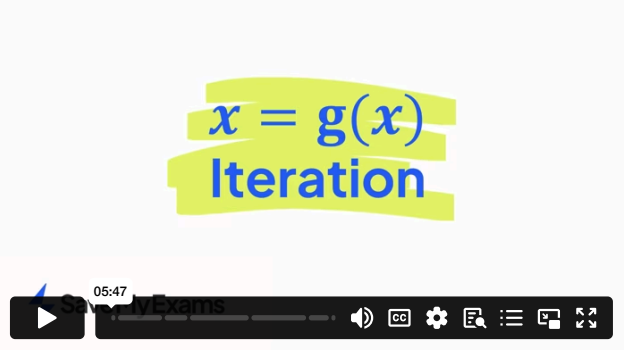
Updated on 23 June 2025

Help me understand



Did this video help you? Yes No

x = g(x) iteration

What is iteration?

- When an equation cannot be solved using the usual **analytical methods**, we can still find **approximate solutions** to a certain degree of accuracy
- **Iteration** is one way to do this, by repeatedly using each answer as the new starting value for a function, we can achieve an ever more accurate answer

What does $x_{n+1} = g(x_n)$ mean?

- Iterations are shown using the notation $x_{n+1} = g(x_n)$
- This is a **recurrence relation** where, starting with a number (x_n), we will get an answer x_{n+1} which we can then reuse in the original function
- Equations need to be rearranged into an **iterative formula** – ie. the form x = g(x)

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

WHEN REARRANGING AN EQUATION INTO AN ITERATIVE FORMULA $g(x)$, PAY ATTENTION TO THE "SHOW THAT" PART OF THE QUESTION TO ENSURE YOU REARRANGE IT INTO THE CORRECT FORMAT

SHOW THAT $x^3 - x^2 - 9 = 0$

CAN BE REARRANGED INTO $x = \sqrt{\frac{9}{x-1}}$

$$x^3 - x^2 - 9 = 0$$

SEPARATING LIKE TERMS CAN BE A GOOD PLACE TO START AS 9 IS ON ITS OWN IN THE ITERATIVE FORMULA

$$\Rightarrow x^3 - x^2 = 9$$

FACTORISING OFTEN HELPS, ALSO THE QUESTION HAS A $\sqrt{\text{SQUARE ROOT}}$ SO GETTING x^2 ON ITS OWN IS HELPFUL

$$x^2(x-1) = 9$$

$$x^2 = \frac{9}{x-1}$$

DIVIDING BY $x-1$ THAN SQUARE ROOTING BOTH SIDES LEAVES THE APPROPRIATE ITERATIVE FORMULA

$$x = \sqrt{\frac{9}{x-1}}$$

$x^3 - x^2 - 9 = 0$ ALSO REARRANGES TO $\sqrt{x = x^3 - 9}$ SO BE CAREFUL TO GET THE CORRECT FUNCTION

ITERATION

GIVEN A STARTING VALUE x_0, YOU CAN **SUBSTITUTE** EACH ANSWER INTO THE **ITERATIVE FORMULA.**
THE NOTATION $x_1, x_2 \ldots$ IS USED TO **TRACK EACH ITERATION.**
ANSWERS MAY **CONVERGE** ON A PARTICULAR NUMBER OR **DIVERGE.**

USING THE ITERATING FORMULA

$x_{n+1} = \sqrt{\frac{9}{x_n - 1}}$ STARTING $x_0 = 2$

a) FIND x_1 AND x_2

b) SOLVE $x^3 - x^2 - 9 = 0$ TO 2dp

a) $x_1 = \sqrt{\frac{9}{2-1}} = 3$

$x_2 = \sqrt{\frac{9}{3-1}} = 2.121\ldots$

SUBSTITUTE x_0 INTO FORMULA AND REPEAT WITH NEW ANSWER

b) $x_3 = 2.833\ldots$

$x_4 = 2.215\ldots$

CONTINUE...

$x = 2.47$ (2 dp)

CONTINUE THIS PROCESS, THE "ANS" BUTTON ON YOUR CALCULATOR IS VERY HELPFUL FOR THIS $\sqrt{\frac{9}{\text{ANS} - 1}}$

STOP ONCE THE ANSWER IS THE SAME TO TWO DECIMAL PLACES WHEN ROUNDED

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What are staircase diagrams and cobweb diagrams?

- Iterations can be shown on diagrams called **staircase** or **cobweb** diagrams
- These can be drawn by **plotting the graphs** of **y = x** against **y = g(x)** from your iterative formula

YOUR ITERATIVE FORMULA (POINTS WHERE THE TWO GRAPHS MEET ARE ROOTS)

STEP 2: MARK x_0 ONTO THE x-AXIS AND DRAW A **VERTICAL LINE** TO MEET THE **CURVE** $y = g(x)$

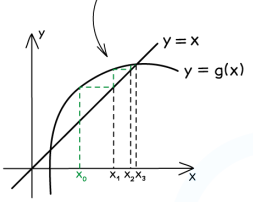
STEP 3: FROM THIS POINT DRAW A **HORIZONTAL LINE** TO MEET THE **LINE** $y = x$. THIS IS x_1.

STEP 4: CONTINUE BY DRAWING ANOTHER **VERTICAL LINE** TO THE **CURVE** $y = g(x)$ THEN A HORIZONTAL LINE TO THE LINE $y = x$. EACH TIME YOU REPEAT THIS STEP IS **ANOTHER ITERATION.**

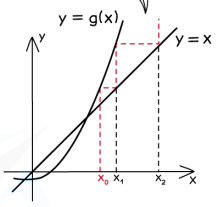
STEP 5: IF THE LINES ARE GETTING **CLOSER** TO THE ROOT THE ITERATION IS **CONVERGING,** IF THEY ARE GETTING **FURTHER** AWAY IT IS **DIVERGING.**

STAIRCASE DIAGRAMS

CONVERGING



DIVERGING

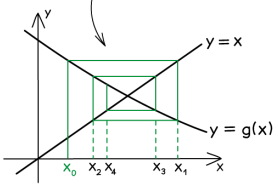


FOR **CONVERGING DIAGRAMS,** EACH ITERATION FROM x_0 WILL GET **CLOSER TO THE ROOT** (POINT WHERE THE TWO GRAPHS INTERSECT)
FOR COBWEB DIAGRAMS THE SOLUTIONS WILL ALTERNATE ABOVE AND BELOW THE ROOT BUT THEY WILL STILL BE GETTING CLOSER EACH TIME

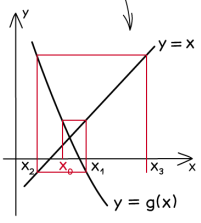
FOR **DIVERGING DIAGRAMS,** EACH ITERATION FROM x_0 WILL GET **FURTHER FROM THE ROOT**

COBWEB DIAGRAMS

CONVERGING



DIVERGING



IN GENERAL THE GRADIENT OF $g(x)$ MUST BE BETWEEN -1 AND 1

IF THE GRADIENT OF $g(x)$ IS **NEGATIVE** AT THE ROOTS IT WILL CREATE A **COBWEB** DIAGRAM

IF THE GRADIENT OF $g(x)$ IS **POSITIVE** AT THE ROOTS IT WILL CREATE A **STAIRCASE** DIAGRAM

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Examiner Tips and Tricks

- You must show all your steps when rearranging an equation into an iterative formula

- Use ANS button on your calculator to calculate repeated iterations
- Keep track of your iterations using x_2, x_3... notation
- Iteration may be part of bigger numerical methods questions

Worked Example

Teacher only

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